

Chapter 1. Review of Probability

1.1 Events and Probability

In this course, we study the probability in the sense of real analysis, which relies on the measure theory. Hence, we will translate the notions of real analysis to the language of probability theory. Now we start with some definitions.

Definition 1.1. Let $\Omega \neq \emptyset$. A family of

$$\mathcal{F} \subset 2^\Omega = \mathcal{P}(\Omega)$$

is a σ -field if it satisfies the following properties:

1. $\emptyset \in \mathcal{F}$.
2. If $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$.
3. If $A_1, A_2, A_3, \dots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

Note that σ -field is nothing but a collection of subsets¹ that is closed under the complement and countable union, which contains the empty set. From now on, we will denote $\mathcal{F}$ as a σ -field on Ω . Based on the above definition, we can easily observe the following remarks.

Remark. Ω and $\mathcal{F}$ are defined as above.

1. $\Omega \in \mathcal{F}$. Use the first two properties to show this.
2. $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$. Use the second and third properties to show this.
3. If $A, B \in \mathcal{F}$, then $A \setminus B \in \mathcal{F}$.

Example. The example of σ -fields are as follows.

1. $2^{\mathbb{R}}$ is a σ -field.
2. (Borel σ -field) Recall that we have defined the Borel σ -field $\mathcal{B}(\mathbb{R})$ in the real analysis class. We say that $\mathcal{B}(\mathbb{R})$ is **Borel σ -field** which is the smallest σ -field containing all open sets² in $\mathbb{R}$. Its elements are called **Borel sets**.
3. The family of (Lebesgue) measurable sets $\mathcal{M}(\mathbb{R})$ is a σ -field.
4. (Coin Tossing) To see more practical examples, consider the following example. For two coin tosses, take

$$\Omega = \{HH, HT, TH, TT\}.$$

If only the first coin toss is observed, let

$$A = \{HH, HT\}, \quad B = \{HH, TH\}.$$

Then the available information is described by the σ -field,

$$\mathcal{F} = \{\emptyset, A, \Omega \setminus A, \Omega\}.$$

Definition 1.2. A **probability measure** is a map

$$P : \mathcal{F} \rightarrow [0, 1]$$

such that

¹That is, the power set of underlying set Ω .

²For the real line $\mathbb{R}$, all open sets mean all open intervals of $\mathbb{R}$.

1. $P(\Omega) = 1$.
2. (Countable Additivity) For pairwise disjoint sets $A_n \in \mathcal{F}$,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

We call Ω the **sample space**, $\mathcal{F}$ the **event space**, and $A \in \mathcal{F}$ the **event**. The triple $(\Omega, \mathcal{F}, P)$ is called a **probability space**. An event A is said to be hold **almost surely (a.s.)** if $P(A) = 1$.

Note that $P(A) = 1$ does not imply that $A = \Omega$.

Example. Take $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$ and $P = \text{Leb}([0, 1])^3$. Indeed,

$$P(\Omega) = P([0, 1]) = \text{Leb}([0, 1]) = 1$$

and more generally, for $[a, b] \subset [0, 1]$,

$$P([a, b]) = \text{Leb}([a, b]) = b - a.$$

Therefore we can observe that such $(\Omega, \mathcal{F}, P)$ is a probability space.

Remark. Let $(\Omega, \mathcal{F}, P)$ be a probability space.

1. $P(\Omega) = 1$.
2. $P(\Omega \setminus A) = 1 - P(A)$ for any $A \in \mathcal{F}$.
3. $A \subset B \implies P(A) \leq P(B)$.
4. (Countable subadditivity) $P(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} P(A_n)$.

The countable *subadditivity* shows that even though we do not have *pairwise* disjoint sets, we still get *inequality*.

Let us define some notations about limit of sets.

If $E_j \subset E_{j+1}$ and $\bigcup_{j=1}^{\infty} E_j = E$, we write $E_j \nearrow E$.

If $E_j \supset E_{j+1}$ and $\bigcap_{j=1}^{\infty} E_j = E$, we write $E_j \searrow E$.

For the measurable sets, we studied that the following corollary holds. Note that $m(\cdot)$ denotes the Lebesgue measure on $\mathbb{R}$ for this moment.

Corollary 3.3 (From Stein's book). Let $E_j \in \mathcal{M}$ for all $j \in \mathbb{N}$.

1. If $E_j \nearrow E$, then $\lim_{j \rightarrow \infty} m(E_j) = m(E)$.
2. If $E_j \searrow E$ and $m(E_k) < \infty$ for some k , then $\lim_{j \rightarrow \infty} m(E_j) = m(E)$.

Similarly, we can obtain the followings for the probability measure P but see the difference with the Lebesgue measure.

Exercise 1.1 (Continuity of Probability). Let $A_i \in \mathcal{F}$.

1. If $A_n \nearrow A$, then $\lim_{n \rightarrow \infty} P(A_n) = P(A)$.
2. If $A_n \searrow A$, then $\lim_{n \rightarrow \infty} P(A_n) = P(A)$.

Remarkably, we can drop the finite condition in the second part, which is different from the Lebesgue measure since $P(\Omega) = 1 < \infty$.

³Here, $\text{Leb}([0, 1])$ denotes the Lebesgue measure on $[0, 1]$.

Again, let us compare the Borel-Cantelli lemma for the Lebesgue measure and the probability measure. In the case of the Lebesgue measure, this lemma implies that if we sum up countably many sets, and if its sum is finite, then the *tail* of the sets have measure zero. Similar happens for the probability measure, which is stated as follows.

Exercise 1.6.16 (The Borel-Cantelli Lemma From Stein's book). Suppose $\{E_k\}_{k=1}^{\infty}$ is a countable family of measurable subsets of $\mathbb{R}^d$ and that

$$\sum_{k=1}^{\infty} m(E_k) < \infty.$$

Let

$$E = \{x \in \mathbb{R}^d : x \in E_k \text{ for infinitely many } k\} = \limsup_{k \rightarrow \infty} E_k.$$

Then, $E \in \mathcal{M}$ and $m(E) = 0$.

Lemma 1.1 (Borel-Cantelli). If $\sum_{n=1}^{\infty} P(A_n) < \infty$ and $B_n = \bigcup_{k \geq n} A_k$, then

$$P\left(\bigcap_{n \geq 1} B_n\right) = 0.$$

Proof. Note that

$$P(B_n) \leq \sum_{k \leq n} P(A_k) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Since $B_n \searrow \bigcap_{n \geq 1} B_n$, we have

$$P\left(\bigcap_{n \geq 1} B_n\right) = \lim_{n \rightarrow \infty} P(B_n) = 0.$$

□

1.2 Random Variables

Start with the definition of random variables.

Definition 1.3. A function $\xi : \Omega \rightarrow \mathbb{R}$ is **$\mathcal{F}$ -measurable**⁴ if for all Borel sets $B \in \mathcal{B}(\mathbb{R})$,

$$\{\xi \in B\} := \{\omega \in \Omega : \xi(\omega) \in B\} = \xi^{-1}(B) \in \mathcal{F}.$$

If $(\Omega, \mathcal{F}, P)$ is a probability space, then such a map ξ is a **random variable**.

Definition 1.4. The σ -field generated by random variable ξ is defined by

$$\sigma(\xi) := \{\{\xi \in B\} : B \in \mathcal{B}(\mathbb{R})\}.$$

Note that $\sigma(\xi)$ contains all the information from observing ξ .

Definition 1.5. For a family $\{\xi_i : i \in \mathcal{I}\}$,

$$\sigma(\xi_i : i \in \mathcal{I})$$

is the smallest σ -field containing every event $\{\xi_i \in B\}$.

⁴Note : Let $E \subset \mathbb{R}^d$. A function $f : E \rightarrow \mathbb{R}^d$ is measurable if for all $a \in \mathbb{R}$, the set

$$\{f < a\} = f^{-1}(\{x \in \mathbb{R} : x < a\}) = \{x \in E : f(x) < a\} \in \mathcal{M}.$$

Thus the measurability is closely related to the inverse image of the function.

Exercise 1.3. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is Borel (measurable)⁵, then $f(\xi)$ is $\sigma(\xi)$ -measurable.

Proof. Trivial. □

Surprisingly, the converse of the Exercise 1.3. is also true. This work is done by Doob and Dynkin, the great mathematicians who contributed to the development of probability theory.

Lemma 1.2 (Doob-Dynkin). *If η is $\sigma(\xi)$ -measurable, then*

$$\eta = f(\xi) \text{ for some Borel function } f : \mathbb{R} \rightarrow \mathbb{R}.$$

To prove this lemma, we have to construct a such Borel function f . The construction of such f is highly non-trivial, thereby we omit the proof of this lemma. It is enough to see the statement for this time.

Definition 1.6. Every random variable ξ gives rise to a probability measure

$$P_\xi(B) = P(\{\xi \in B\}) \quad \text{for } B \in \mathcal{B}(\mathbb{R}).$$

P_ξ is called the **distribution** of ξ . The function $F_\xi(x) : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_\xi(x) = P_\xi((-\infty, x]) = P(\{\xi \leq x\})$$

is called the **distribution function** of ξ .

Exercise 1.4. F_ξ is non-decreasing, right-continuous, and satisfies

$$\lim_{x \rightarrow -\infty} F_\xi(x) = 0, \quad \text{and} \quad \lim_{x \rightarrow \infty} F_\xi(x) = 1.$$

Proof.

1. Monotonicity is immediate.
2. If $x_n \searrow x$, then $\{\xi \leq x_n\} \searrow \{\xi \leq x\}$ hence $F_{x_n} \rightarrow F_\xi(x)$.
3. For the limits, use

$$\{\xi \leq -n\} \searrow \emptyset \quad \text{and} \quad \{\xi \leq n\} \nearrow \Omega.$$

□

Definition 1.7. The law of ξ is **absolutely continuous**⁶ if there is a Borel function $f_\xi : \mathbb{R} \rightarrow \mathbb{R}$ such that for any Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$P_\xi(B) = P(\xi \in B) = \int_B f_\xi(x) dx$$

and f_ξ is called the **density** of ξ . The law is **discrete** if there exists at most countable pairwise distinct values $\{x_i\}_{i \in \mathcal{I}}$ such that

$$P(\xi \in B) = \sum_{x_i \in B} P(\xi = x_i) \quad \text{for all } B \in \mathcal{B}(\mathbb{R}).$$

Exercise 1.5 and 1.6. For a continuous density f_ξ , we have

$$F'_\xi(x) = f_\xi(x).$$

For a discrete law,

$$F_\xi \text{ jumps by } P(\xi = x_i) \text{ at each } x_i$$

⁵Our textbook defines **Borel function** as follows. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is Borel if the inverse image $f^{-1}(B)$ of any Borel set B is a Borel set. In other words, f is Borel if it is measurable with respect to the Borel σ -field.

⁶That is, the distribution of ξ is absolutely continuous.